



Introduction

Bipolar disorder (BD) is strongly associated with financial instability [1]. Symptomatic periods in BD often manifest in poor financial decision-making [2]. For example, 70% individuals with BD have reported impulsive spending during hypomania [3]. Such problematic financial behaviors can lead to serious long-term financial instability, which can severely impact the quality of life for individuals with BD [1] and their care partners [4].

However, assessment of financial behavior is often limited to self-report. This study (N=50) used open banking technologies to collect 24 months of transaction data and monthly self-report mood logs from individuals with bipolar disorder.

Methods

Participants from the U.S. were recruited via Depression and Bipolar Support Alliance and NIH ResearchMatch between July 2024 and August 2025.

Participants reviewed journals and photographs to aid in symptom recall. Monthly severity ratings of mania and depression were collected on a scale of 0 (euthymic) to 3 (hospitalized or incapacitated) via the NIMH Life Chart Method Retrospective (LCM-R). Participants interacted with a secured web application to retrieve de-identified financial data from their financial institutions using Plaid, Inc.'s financial data aggregation service.

Results

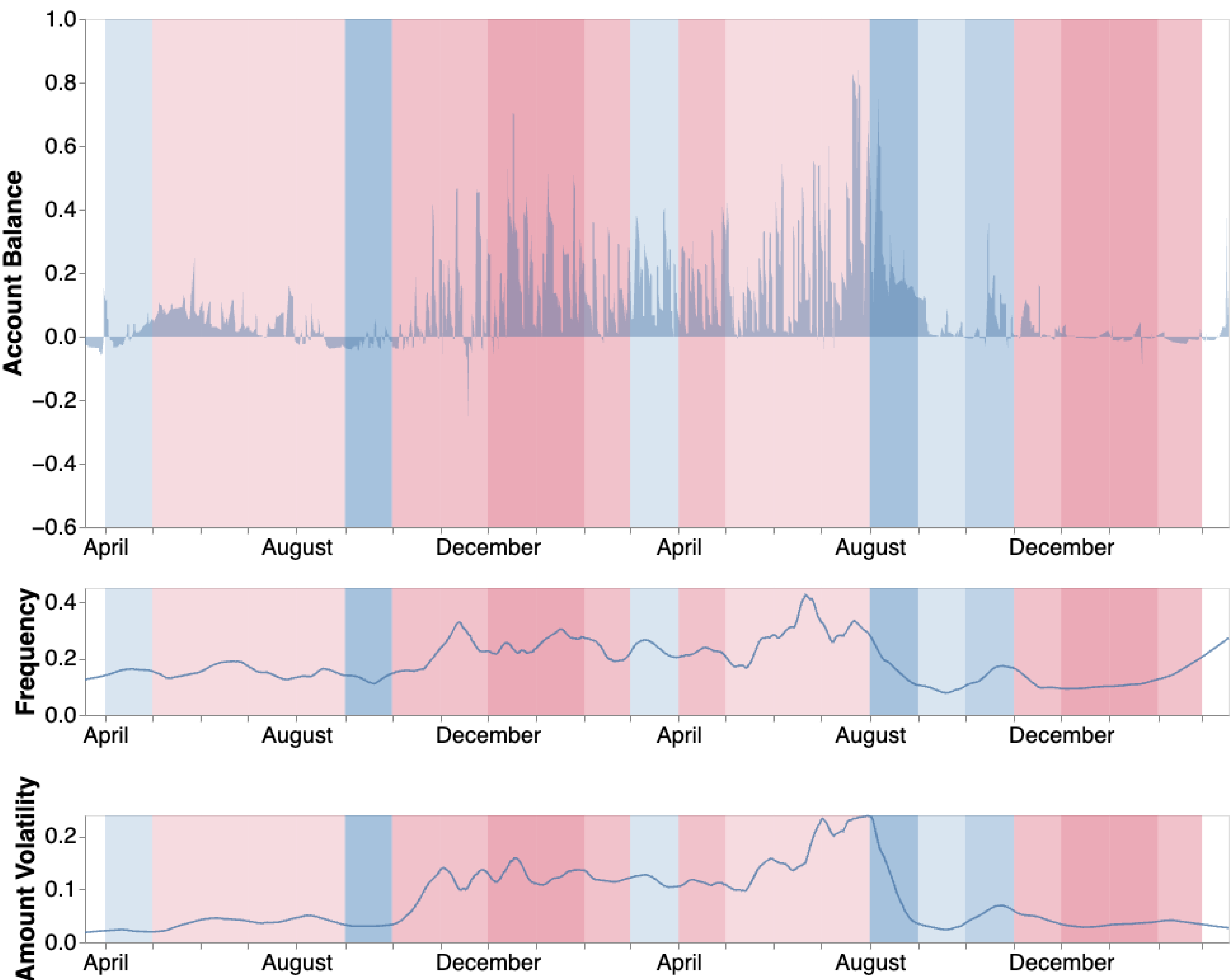
In our sample, 10 identified as male, 36 as female, 3 as non-binary, and 1 did not disclose. 21 participants were diagnosed with BD-I, 23 with BD-II, 3 with BD Not Otherwise Specified, and 3 were unsure.

This dataset contains 67,668 categorized transactions across 890 total days.

Transactions are annotated with severity ratings of mania and depression.

Preliminary analysis via changepoint trend detection and Hidden Markov models suggest financial data are predictive of latent illness state.

This material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE1255832 and by the National Institutes of Health's National Institute of Mental Health under award number R21MH131924. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation. Approved by Pennsylvania State University IRB, STUDY00019759.



Normalized account balance (top), daily transaction frequency (middle), and transaction amount volatility (bottom, 14-day rolling standard deviation) for a single individual. Mood symptoms (mania in red, depression in blue) reflect NIMH LCM-R ratings with darker shading representing increased mood severity. Transaction frequency and volatility show a marked increase during hypomania.

Discussion

The lack of access to objective data has been a key barrier in establishing the relationship between financial stability and BD. This research aims to help close the knowledge gap regarding how idiosyncratic, context-driven, and illness-specific factors impact financial decision-making and stability in BD.

Further, the lack of granular, in-situ assessment methods is a key challenge against developing just-in-time and personalized interventions focusing on financial stability for this population.

Future work can extend this research direction by developing digital health interventions which incorporate therapeutic modalities (e.g., Acceptance and Commitment Therapy, Cognitive Financial Behavior Therapy) and granular financial data in order to support financial stability during symptomatic and euthymic states.

Conclusions

This study demonstrates the feasibility of collecting granular, longitudinal financial data in this population, providing an objective alternative to self-report for assessing illness-related financial instability.

This dataset will be published in the NIMH Data Archive later this year.

References

[1] Richardson, T., Jansen, M., & Fitch, C. (2021). Financial difficulties in bipolar disorder part 2: Psychological correlates and a proposed psychological model. *Journal of Mental Health*, 30(1), 3–11.

[2] Ramírez-Martín, A., Ramos-Martín, J., Mayoral-Cleries, F., Moreno-Küstner, B., & Guzman-Parra, J. (2020). Impulsivity, decision-making and risk-taking behaviour in bipolar disorder: A systematic review and meta-analysis. *Psychological Medicine*, 50(13), 2141–2153.

[3] Fletcher, K., Parker, G., Paterson, A., & Synnott, H. (2013). High-risk behaviour in hypomanic states. *Journal of Affective Disorders*, 150(1), 50–56.

[4] Beentjes, T. A. A., Goossens, P. J. J., & Poslawsky, I. E. (2012). Caregiver Burden in Bipolar Hypomania and Mania: A Systematic Review. *Perspectives in Psychiatric Care*, 48(4), 187–197.